



INFO

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AREAS

Mobile robotics · Marine robotics ·
Motion planning · Decision making
· Obstacle detection and tracking ·
Control systems

TOOLS

ROS · ROS 2 · Nav2 · Gazebo · ArduPilot
· ArduSub · QGroundControl · LiDAR ·
Codesys · Modbus TCP · OPC-UA · MATLAB
· Onshape · Docker · Linux · Git · LaTeX

LANGUAGES

C++ · C · Python · Java

SPOKEN

Portuguese (native)
English (fluent)
German (beginner)

Francisco Pires da Silva

Robotics Engineer | Mobile & Marine Robotics

Experience

Research Assistant at INESC TEC, Centre for Robotics and Autonomous Systems (CRAS)

Motus Robotics group 

Jan 2023 – Sep 2026 | Porto, Portugal

Worked on robotic vehicle navigation for offshore inspection across three EU-funded projects, on motion planning, obstacle detection and avoidance, control architecture and decision making. Accumulated 30+ days of field experience across trials and live demonstrations.

AEROSUB | Horizon Europe · automated offshore wind inspection

- Dynamic collision avoidance for close-range inspection, extending the velocity-obstacle work from the MSc dissertation to ASV interaction on offshore inspection routines.

TALOS | Horizon Europe · robotics and AI for photovoltaic inspection

- Developed the full software stack for the **LILY ASV**: motion planning, decision making, control architecture and UAV coordination.
- Demonstrated the fully autonomous mission at Barragem de Alqueva, Portugal .

ATLANTIS | H2020 · Atlantic offshore wind test platform

- Conceptualised and prototyped an active docking and retention system for UAV launch and recovery at sea.

Education

PhD Coursework, Electrical and Computer Engineering

Sep 2025 – Aug 2026 | University of Porto (FEUP)

Curricular year completed.

Exchange Semester, Electrical and Computer Engineering

Oct 2024 – Mar 2025 | Technical University of Munich (TUM), Germany

MSc in Electrical and Computer Engineering

Sep 2023 – Jul 2025 | University of Porto (FEUP) | GPA 18/20

Branch: Automation | Profile: Robotics and Autonomous Systems

Dissertation: *A Velocity Obstacle Avoidance Algorithm for Heterogeneous Robotic Vehicles*  (19/20). A 3D velocity-obstacle algorithm that handles different vehicle kinematics under one method, letting holonomic and differential-drive surface vehicles and multirotor UAVs navigate through dynamic obstacles. Validated in simulation and in field trials on the NEST ASV at the ATLANTIS Test Centre.

BSc in Electrical and Computer Engineering

Sep 2020 – Jul 2023 | University of Porto (FEUP) | GPA 18/20

Courses, Training & Certifications

MIT Portugal Marine Robotics Summer School, Azores, Jul 2026

Marine Sensors for Ocean Environments (INESC TEC / SINTEF training course), Porto, Sep 2026

UAV Pilot Licence: EASA Open Category A1/A3, ANAC Portugal

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Leadership & Outreach

NEEEEC-FEUP

Sep 2020 – Aug 2025 | Electrical and Computer Engineering Students' Branch, FEUP

- **Board Member, 2023/24:** head of the technological department, leading a team of 20 on projects and workshops.
- Logistics coordinator for NEEEL-IT, a robotics competition.
- Organised and presented hands-on technical workshops for students on ROS 2, electronics, Arduino and Notion.
- Member of EESTEC, the European Association of Students of Electrical Engineering, since March 2022.

Mentor, FEUP Mentorship Program

2022/23 | University of Porto (FEUP)

Scouting, CNE Agrupamento 212 São Pedro

2008 – Aug 2020 | Vila Real, Portugal

Selected Projects (more details on xizesse.github.io

Bird Recovery Detection (2026 – present): YOLO26 and object-tracking vision pipeline tracking the recovery of rehabilitated birds, for the wildlife recovery centre at CRAS-HVUTAD.

Hull Cleaning ROV (2026, MIT Portugal MRSS): Adapted an ROV for detection and cleaning of invasive species carried on ship hulls, in the Azores.

Christmas Market Methane Study (2024, TUM): Mobile methane emission survey of the Munich Christmas markets, for the Environmental Sensing and Monitoring chair.

Autonomous Quadruped Robot (2023, FEUP) : Walking quadruped on an RP2040, with IMU and time-of-flight obstacle detection.

PID Ball Balancing Robot (2023, NEEEC): Closed-loop control demonstrator built for the University of Porto fair, for high-school outreach.

H.O.W.D.Y. (2023) : Networked multiplayer puzzle game in Java, where players cooperate from separate devices.